



DATA SCIENCE AND ENGINEERING BACHELOR DEGREE

Home > Academics > Faculties and Institutes > Faculty Of Engineering
 > Department of Information Technology Engineering > Data Science And Engineering Bachelor Degree

Data Science And Engineering Bachelor Degree

Introduction to Degree

The environment is a vital element required for human existence and development. However, due to human actions, environmental problems including global warming, climate change, pollution, natural resources depletion and hazardous waste disposal occur, and their impacts are aggravated by human-related activities and decisions. To manage the problems, the most relevant approach or technique is required to be selected and applied, based on results from environmental assessment and monitoring at the local, national and international levels. The impacts of the environmental problems are more serious in developing countries like Cambodia compared to developed countries due to limitations of knowledge and human capacity as well as lack of financial availability to manage them. International efforts to advance environmental conservation in developing countries have been carried out and contributed to increasing the quality of life for people and societies in the countries. However, it is now recognized that the capacity of human resources and academic institutes needs to be developed and strengthened for environmental management and conservation by the countries themselves. The Department of Environmental Engineering provides an Honor Bachelor Program designed for exceptionally motivated and high-achieving students who want thorough, in-depth education about environmental challenges and solutions. Through the top-notch program, the Department with support from the Korea International Cooperation Agency (KOICA) and the Gwangju Institute of Science and Technology (GIST), based in the Republic of Korea, educates students to understand interactions between human actions and environmental effects. Moreover, the Department develops the capability of students to identify and address environmental challenges by adopting widely used or state-of-the-art approaches and technologies.

Last semester Optional tracks



Course Work	is for students who would like to conduct coursework and final project.
Research Thesis	is for outstanding student who would like to conduct their research project under the supervision of BE faculty members.
Extended Training	is designed for students who would like to participate in a long training period (for the entire semester) under a co-operative training program with companies or organizations.

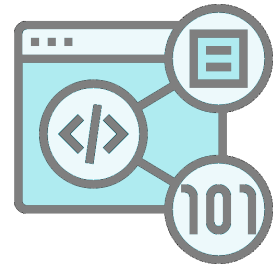
Prospects

The Department carries out research in which environmental impact and risk are assessed, monitored and/or address for enhancing environmental conservation and sustainability in Cambodia. Findings from the research can be used to facilitate recovering the environment and develop solutions to minimize the impacts on and risk to environmental resources in the country

Research knowledge which are provided by Data Science Engineering department

Research

- Web and Cloud Technology
- Artificial Intelligence or Machine Learning
- Big Data
- Business Intelligence



Based on knowledge which are provided by Data Science Engineering department, graduates can work as



Job Prospects

- Data Analyst
- Business Intelligent Analyst
- Associate Data Engineer
- Data Engineer
- AI/ML Engineer

Evaluation & Graduation

Assessment of study will be performed at the end of each semester. Evaluation will be shown as the Grade Point Average (GPA) (4.00 scale) and the Cumulative Grade Point Average (4.00 scale).

Score Range	GPA	Grade Meaning
85–100	4.0	A = Excellent
80–84	3.5	B+ = Very Good
70–79	3.0	B = Good
65–69	2.5	C+ = Fairly Good
50–64	2.0	C = Fair
45–49	1.5	D = Poor
<40	0.0	E = Fail

Curriculum

The curriculum for the Honor Bachelor Program is designed with the aim of educating fundamental and principal knowledge and skills in environmental engiThe curriculum of Data Science & Engineering Department is designed for our students with abilities to use new technologies and

theories to design and develop AI and Machine Learning

L = Lecture, P = Practice, S = Self Study

Foundation

Subject	Credit	Hours (L - P - S)
SEMESTER I		
English Composition	3.0	(6-0)
Mathematics I (Algebra)	3.0	(3-0)
Physics I	3.0	(2-2)
General Chemistry	3.0	(2-2)
Computer Fundamental and programming	3.0	(2-2)
Introduction to Engineering	3.0	(3-0)
Personal Development and Management	3.0	(3-0)
Digital skills and problems solving	3.0	(3-0)
SEMESTER II		
English: Introduction to Technical and Business Writing	3.0	(6-0)
Mathematics II (Calculus I)	3.0	(3-0)
Biological & Environmental Engineering	3.0	(3-0)
Engineering Graphics	3.0	(2-2)
Introduction to Data Science and Engineering	2.0	(3-0)
Computer Programming Language	3.0	(2-2)
Cambodia and World cultures	3.0	(3-0)
Statistics for Engineers	3.0	(2-2)

Second Year

Subject	Credit	Hours (L - P - S)
SEMESTER I		
Math III: Calculus II	3.0	(3-0)
Probability and Statistics	3.0	(3-0)
Data Structures and Algorithms	3.0	(3-0)
Object-oriented Design and Programming	3.0	(2-2)
Computer Network and Telecommunications	3.0	(3-0)
Operating Systems	3.0	(2-2)
Project Practicum	2.0	(0-4)
SEMESTER II		

Subject	Credit	Hours (L - P - S)
Discrete Math	3.0	(3-0)
Statistics	3.0	(2-2)
Database Design and Management	3.0	(2-2)
Software Engineering	3.0	(3-0)
Web and Cloud Technology	3.0	(2-2)
Introduction to Distributed Computing	3.0	(3-0)
Project Practicum	2.0	(0-4)

Third Year

Subject	Credit	Hours (L - P - S)
SEMESTER I		
Introduction to Big Data and Web Mining	3.0	(2-2)
Artificial Intelligence	3.0	(3-0)
Data Analytics and Machine Learning	3.0	(2-2)
Project Management	3.0	(3-0)
Mobile Application Technology	3.0	(2-2)
Project Practicum	2.0	(0-4)
SEMESTER II		
Optimization and Process Analytics	3.0	(3-0)
Cognitive Computing	3.0	(2-2)
Deep Learning and Application	3.0	(2-2)
Innovation and Entrepreneurships	3.0	(3-0)
Introduction to Cryptography	3.0	(3-0)
Project Practicum	2.0	(0-4)

Fourth Year

Subject	Credit	Hours (L - P - S)
SEMESTER I		
Professional Computing and Ethics	3.0	(3-0)
Marketing Analytics	3.0	(2-2)
Supply Chain Analytics	3.0	(3-0)
Information Security	3.0	(3-0)
Introduction to Enterprise Information Systems	3.0	(3-0)
Project Practicum	2.0	(0-4)
SEMESTER II		
Leadership Skill	3.0	(3-0)

Subject	Credit	Hours (L - P - S)
Professional Internship	8.0	(0-0)
Final Project	7.0	(0-0)

About RUPP

The Royal University of Phnom Penh is Cambodia's oldest and largest public university, offering degrees in science, technology, humanities, and social sciences.

Quick Links

- ACADEMICS
- FACULTIES AND INSTITUTE
- RESEARCH AND PROJECTS
- E-LIBRARY

Contact Us

Russian Federation Blvd, Phnom Penh, Cambodia
info@rupp.edu.kh
+855 23 883 640

Connect With Us

Stay updated with our latest news and events